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AID Semester 2

LAB EXCERCISE 8

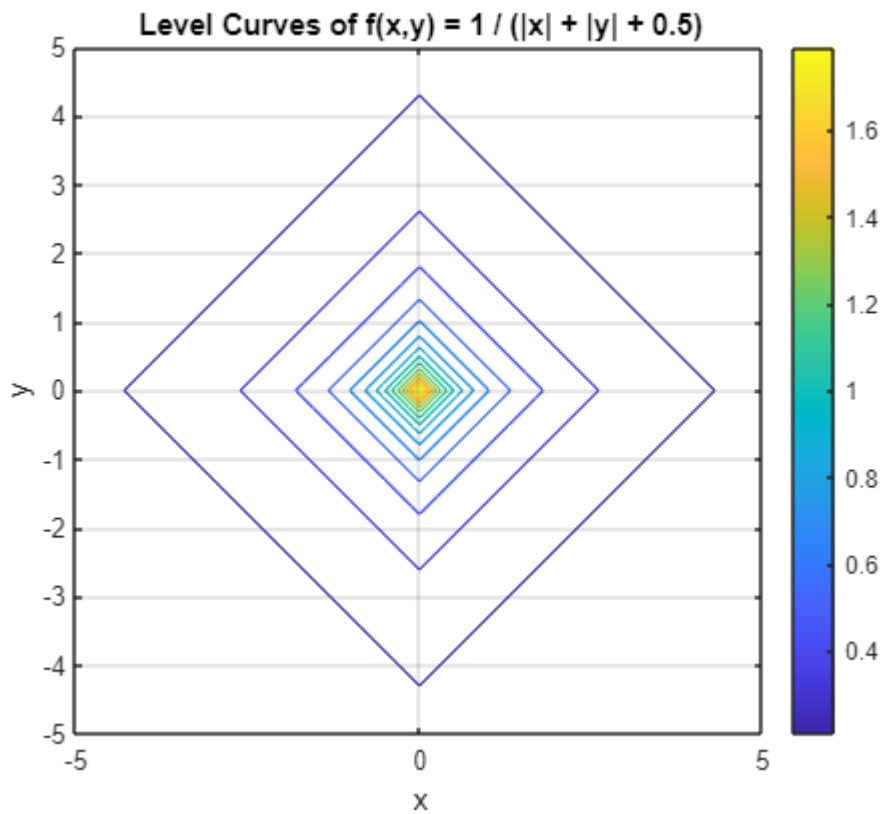
1. Consider the function: f(x,y) = 1/(|x| + |y| + 0.5). Plot 15 level curves for the function f(x,y) in the rectangular domain $-5 \le x,y \le 5$.

```
x=linspace(-5,5,400);
y=linspace(-5,5,400);
[X,Y]=meshgrid(x,y);

%Defining function
Z=1./(abs(X)+abs(Y)+0.5);

%plotting 15 level curves
figure;
contour(X,Y,Z,15);
colorbar;%gives 15 level curves

%labels and titles
xlabel('x');
ylabel('y');
title('Level Curves of f(x,y) = 1 / (|x| + |y| + 0.5)');
axis equal;
grid on;
```



2. Consider the function: f(x,y) = 4x^2 + 9y^2 – 72.

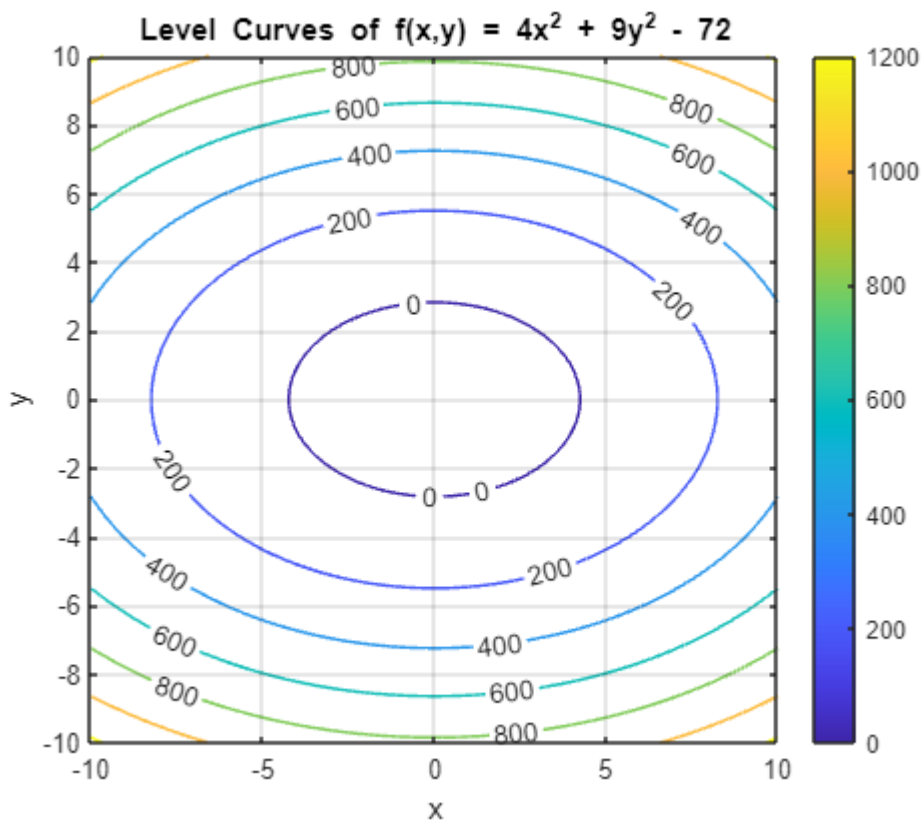
(a) Plot the level curves for f(x,y) with function values written in each curve in $-10 \le x,y \le 10$

```
% Domain
x = linspace(-10, 10, 400);
y = linspace(-10, 10, 400);
[X, Y] = meshgrid(x, y);

% Function
Z = 4*X.^2 + 9*Y.^2 - 72;

% Plot level curves with labels
figure;
[C, h] = contour(X, Y, Z);
clabel(C, h);
colorbar;

xlabel('x');
ylabel('y');
title('Level Curves of f(x,y) = 4x^2 + 9y^2 - 72');
axis equal;
grid on;
```



(b) Find the gradient and hessian of the function using ‘syms’.

```
syms x y

f = 4*x^2 + 9*y^2 - 72;

grad_f = gradient(f, [x, y]);

H = hessian(f, [x, y]);

disp('Gradient:');
```

```
Gradient:
```

```
disp(grad_f);
```

$$\begin{pmatrix} 8\,x \\ 18\,y \end{pmatrix}$$

```
disp('Hessian:');
```

```
Hessian:
```

```
disp(H);
```

$$\begin{pmatrix} 8 & 0 \\ 0 & 18 \end{pmatrix}$$

(c) Plot the gradient vectors at some points on the level curves.

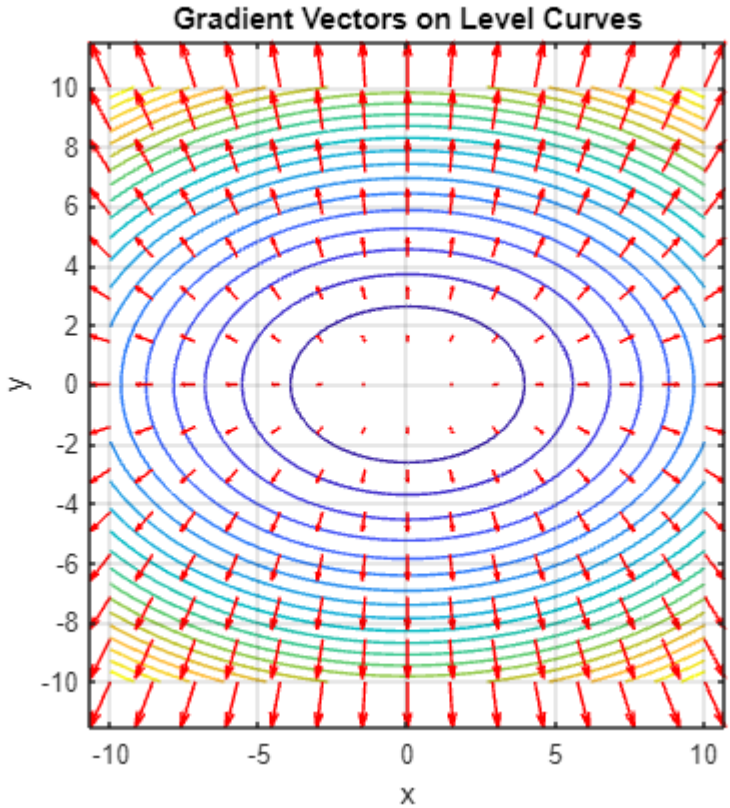
```
% Grid for vector field (coarser grid for clarity)
xg = linspace(-10, 10, 15);
yg = linspace(-10, 10, 15);
[Xg, Yg] = meshgrid(xg, yg);

% Gradient components
U = 8 * Xg;
V = 18 * Yg;

% Plot contours
figure;
contour(X, Y, Z, 20);
hold on;

% Plot gradient vectors
quiver(Xg, Yg, U, V, 'r');

xlabel('x');
ylabel('y');
title('Gradient Vectors on Level Curves');
axis equal;
grid on;
hold off;
```



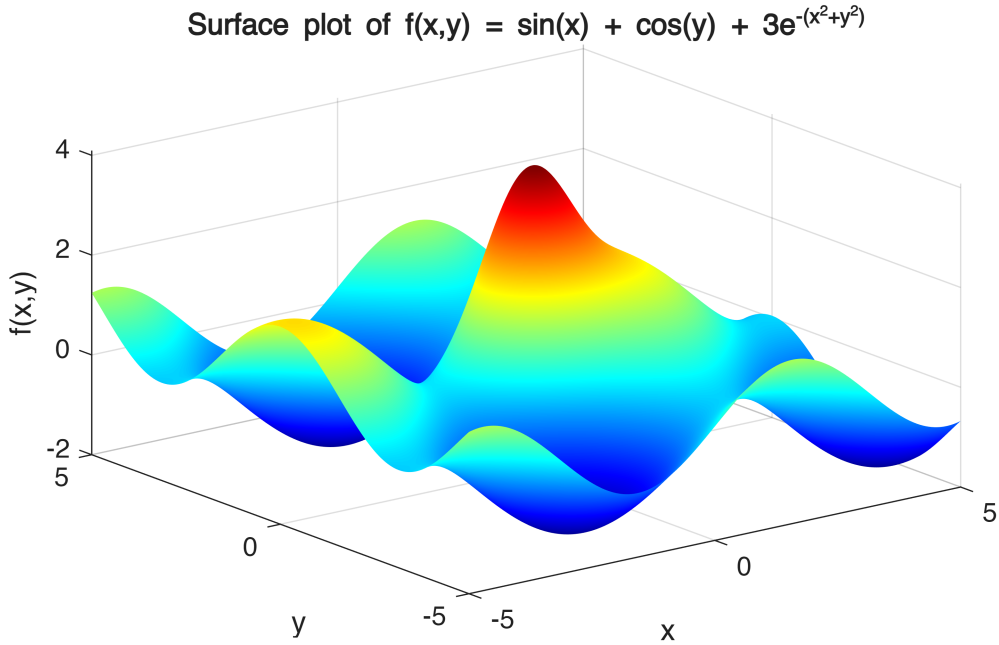
3. Consider the function f(x,y) = sinx + cosy + 3e^(–x2–y2).

(a) Show the surface plot of the function in one figure and the gradient vectors at some points on level curves in another figure.

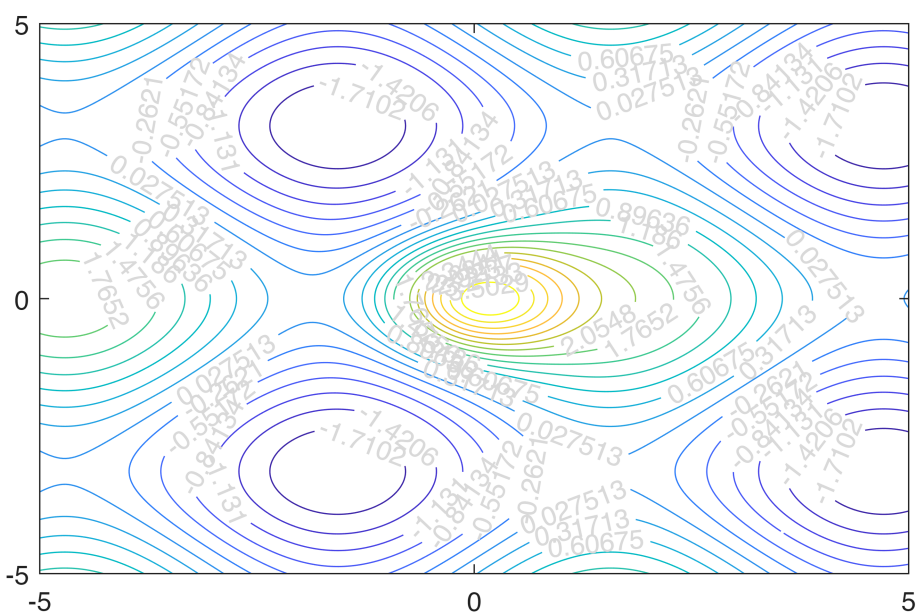
```
% Domain
x = linspace(-5,5,300);
y = linspace(-5,5,300);
[X,Y] = meshgrid(x,y);

% Function
Z = sin(X) + cos(Y) + 3*exp(-X.^2 - Y.^2);

%% ---- Surface plot ----
figure
surf(X,Y,Z)
shading interp
colormap jet;
xlabel('x'); ylabel('y'); zlabel('f(x,y)')
title('Surface plot of f(x,y) = sin(x) + cos(y) + 3e^{-(x^2+y^2)}')
grid on
```




```
%% ---- Contour + Gradient vectors ----
figure
[C,h] = contour(X,Y,Z,20);
clabel(C,h);
```

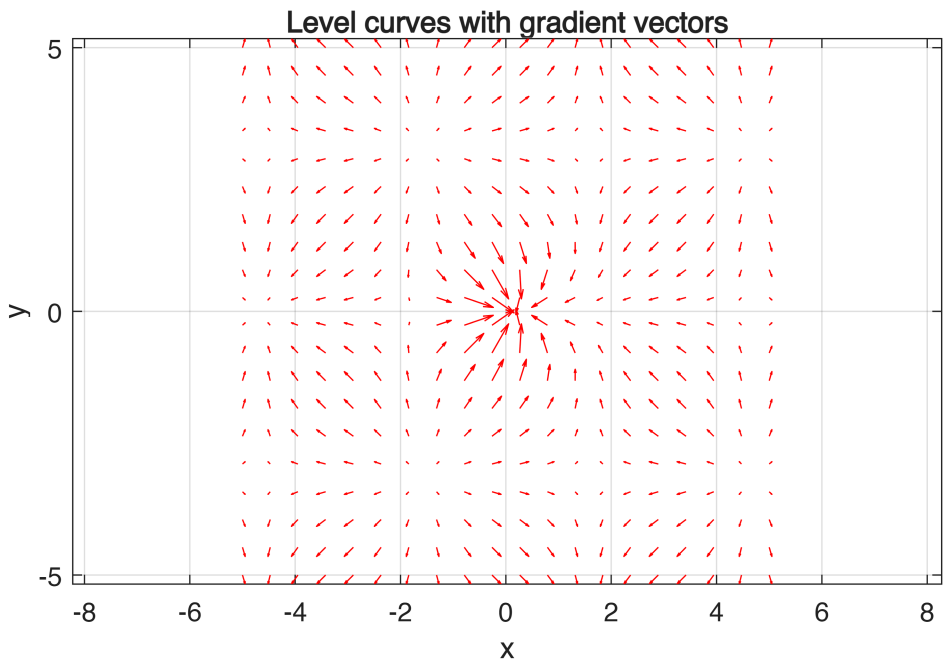



```
% Coarse grid for vectors
xg = linspace(-5,5,20);
yg = linspace(-5,5,20);
[Xg,Yg] = meshgrid(xg,yg);
```

```
% Analytical gradient
U = cos(Xg) - 6*Xg.*exp(-Xg.^2 - Yg.^2);   % df/dx
V = -sin(Yg) - 6*Yg.*exp(-Xg.^2 - Yg.^2); % df/dy

quiver(Xg,Yg,U,V,'r')

xlabel('x'); ylabel('y')
title('Level curves with gradient vectors')
axis equal
grid on
```



(b) Looking at the surface plot and from the figure with function values on level curves, try to find a minimum and a maximum for the function.

```
% Maximum
[max_val, idx_max] = max(Z(:));
x_max = X(idx_max);
y_max = Y(idx_max);

% Minimum
[min_val, idx_min] = min(Z(:));
x_min = X(idx_min);
y_min = Y(idx_min);

% Display
fprintf('Maximum: f = %.4f at (%.4f, %.4f)\n', max_val, x_max, y_max);
```

```
Maximum: f = 4.0821 at (0.1839, -0.0167)
```

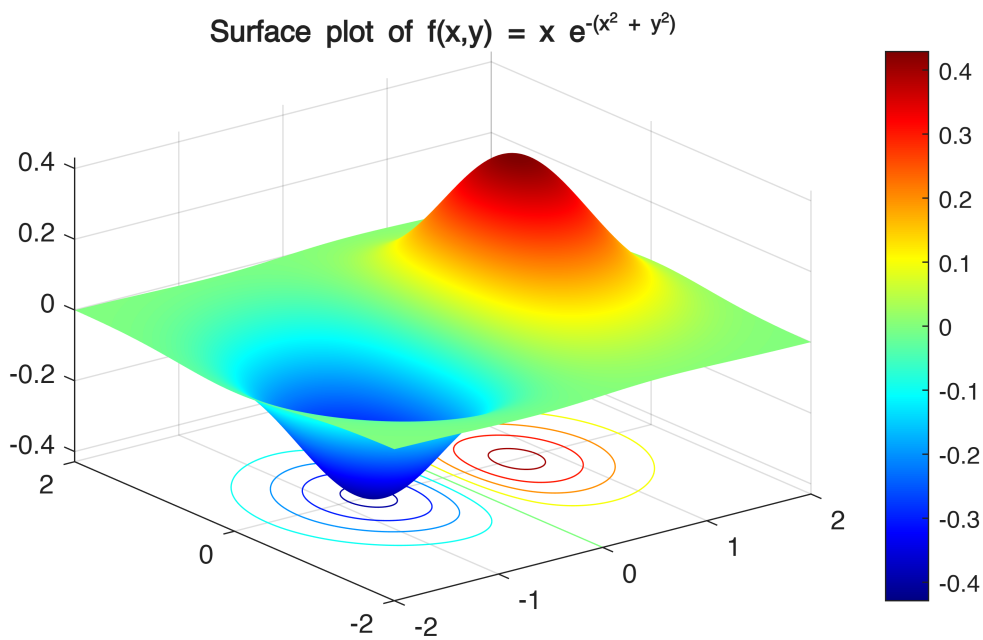
```
fprintf('Minimum: f = %.4f at (%.4f, %.4f)\n', min_val, x_min, y_min);
```

```
Minimum: f = -1.9998 at (4.6990, -3.1271)
```

4. Consider the function: f(x,y) = x e^{-(x^2 + y^2)

a) Plot the surface given for x and y varying from -2 to 2

```
clear;
clc;
close all;
x = linspace(-2,2,200);
y = linspace(-2,2,200);
[X,Y]=meshgrid(x,y);
Z=X.*exp(-(X.^2 + Y.^2));
figure;
surfc(X,Y,Z);
colormap jet;
shading interp;
colorbar;
title('Surface plot of f(x,y) = x e^{-(x^2 + y^2)}')
```



b) Write the General expression for gradient and Hessian for the function

```
syms x y
f = x*exp(-(x^2 + y^2));

grad_f = gradient(f, [x y])
```

```
grad_f =
```

$$\begin{pmatrix} e^{-x^2-y^2} - 2\,x^2\,e^{-x^2-y^2} \\ -2\,x\,y\,e^{-x^2-y^2} \end{pmatrix}$$

```
hess_f = hessian(f, [x y])
```

```
hess_f =
```

$$\begin{pmatrix} 4\,x^3\,\sigma_1 - 6\,x\,\sigma_1 & 4\,x^2\,y\,\sigma_1 - 2\,y\,\sigma_1 \\ 4\,x^2\,y\,\sigma_1 - 2\,y\,\sigma_1 & 4\,x\,y^2\,\sigma_1 - 2\,x\,\sigma_1 \end{pmatrix}$$

where

$$\sigma_1 = e^{-x^2-y^2}$$

c) Evaluate Hessian at the points given in the following figures

```
H1 = subs(H, [x y], [-0.6 0.2])
```

```
H1 =
```

$$\begin{pmatrix} \frac{342\,e^{-\frac{2}{5}}}{125} & -\frac{14\,e^{-\frac{2}{5}}}{125} \\ -\frac{14\,e^{-\frac{2}{5}}}{125} & \frac{138\,e^{-\frac{2}{5}}}{125} \end{pmatrix}$$

```
H2 = subs(H, [x y], [0.6 -0.1])
```

```
H2 =
```

$$\begin{pmatrix} -\frac{342\,e^{-\frac{37}{100}}}{125} & \frac{7\,e^{-\frac{37}{100}}}{125} \\ \frac{7\,e^{-\frac{37}{100}}}{125} & -\frac{147\,e^{-\frac{37}{100}}}{125} \end{pmatrix}$$

```
H3 = subs(H, [x y], [1.6 -0.7])
```

```
H3 =
```

$$\begin{pmatrix} \frac{848\,e^{-\frac{61}{20}}}{125} & -\frac{721\,e^{-\frac{61}{20}}}{125} \\ -\frac{721\,e^{-\frac{61}{20}}}{125} & -\frac{8\,e^{-\frac{61}{20}}}{125} \end{pmatrix}$$

```
H4 = subs(H, [x y], [-1.4 0.35])
```

```
H4 =
```

$$\begin{pmatrix} -\frac{322\,e^{-\frac{833}{400}}}{125} & \frac{511\,e^{-\frac{833}{400}}}{250} \\ \frac{511\,e^{-\frac{833}{400}}}{250} & \frac{1057\,e^{-\frac{833}{400}}}{500} \end{pmatrix}$$

```
H5 = subs(H, [x y], [0 0.2])
```

```
H5 =
```

$$\begin{pmatrix} 0 & -\frac{2\,e^{-\frac{1}{25}}}{5} \\ -\frac{2\,e^{-\frac{1}{25}}}{5} & 0 \end{pmatrix}$$

d) Find eigenvalues of those Hessian matrices and decide the nature of curvature at the points where it is evaluated. (curved upward, downward, neither)

```
eig(H1)
```

```
ans =
```

$$\begin{pmatrix} -\frac{2\,e^{-\frac{2}{5}}\left(\sqrt{106}-24\right)}{25} \\ \frac{2\,e^{-\frac{2}{5}}\left(\sqrt{106}+24\right)}{25} \end{pmatrix}$$

```
eig(H2)
```

```
ans =
```

$$\begin{pmatrix} \frac{e^{-\frac{37}{100}}\left(\sqrt{38221}-489\right)}{250} \\ -\frac{e^{-\frac{37}{100}}\left(\sqrt{38221}+489\right)}{250} \end{pmatrix}$$

```
eig(H3)
```

```
ans =
```

$$\begin{pmatrix} -\frac{e^{-\frac{61}{20}}\left(\sqrt{28121}-84\right)}{25} \\ \frac{e^{-\frac{61}{20}}\left(\sqrt{28121}+84\right)}{25} \end{pmatrix}$$

```
eig(H4)
```

ans =

$$\begin{pmatrix} \frac{7\,e^{-\frac{833}{400}}\left(\sqrt{197489}-33\right)}{1000} \\ -\frac{7\,e^{-\frac{833}{400}}\left(\sqrt{197489}+33\right)}{1000} \end{pmatrix}$$

```
eig(H5)
```

ans =

$$\begin{pmatrix} -\frac{2}{5\,\sqrt{e^{2/25}}} \\ \frac{2}{5\,\sqrt{e^{2/25}}} \end{pmatrix}$$